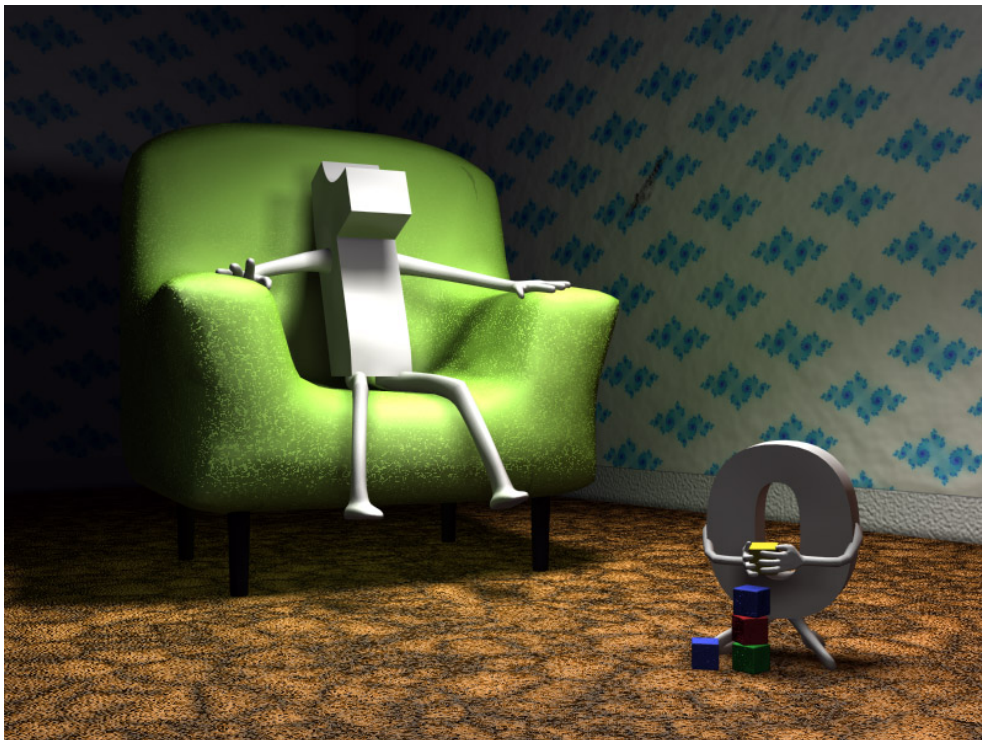


Learning how to prove Week 1

Integers



**Induction. Division Algorithm. Gcd. Primes and coprimes
integers. Euclidean Algorithm.**

1. **Prove each statement by induction on n .**

(a) $\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}} \geq \sqrt{n}$, for all $n \geq 1$.

(b) If $\gcd(a, b) = 1$, prove that $\gcd(a, b^n) = 1$, for all $n \geq 1$.

2. **If n and $d \neq 0$ are integers, show that there exist integers q and r such that $n = qd + r$ and $0 \leq r < |d|$.**